

A 6^r -Scaled Family of the Reduced $6k \pm \{1, 2\}/5$ Collatz-Type Map with Three Observed Attractors

Exact Algebraic Conjugacy, Three-Attractor Transport, Orbit-Statistic Invariance, and Transfer of the 10^{11} Finite Verification

Farhad Banazadeh
fn.bana@hotmail.com
ORCID: 0009-0004-7023-0298

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Abstract

We develop the full 6^r -scaled family of the reduced $6k \pm \{1, 2\}/5$ Collatz-type map previously studied computationally through 10^{11} . The base domain is $\mathcal{D} = \{y \in \mathbb{Z}_{>0} : 5 \nmid y\}$, with correction $c(y) \in \{-2, -1, 1, 2\}$ selected by $y \pmod{5}$, affine numerator $M_0(y) = 6y + c(y)$, and complete 5-adic reduction $T_0(y) = M_0(y)/5^{\nu_5(M_0(y))}$. For every integer $r \geq 0$ we define $\mathcal{D}_r = 6^r \mathcal{D}$ and $M_r(x) = 6x + 6^r c(x)$, whose four branches are $6x - 6^r$, $6x - 2 \cdot 6^r$, $6x + 2 \cdot 6^r$, and $6x + 6^r$. Since $6^r \equiv 1 \pmod{5}$ and $5 \nmid 6^r$, multiplication by 6^r preserves both branch selection and the complete 5-adic valuation. Hence

$$T_r(6^r y) = 6^r T_0(y), \quad T_r^j(6^r y) = 6^r T_0^j(y) \quad (j \geq 0).$$

Thus trajectories, valuation sequences, transient lengths, basin membership, and minimal cycle lengths are transported bijectively; absolute peaks scale by 6^r and normalized peak ratios are invariant. The base fixed points 1, 2 become $6^r, 2 \cdot 6^r$, while the observed 9-cycle becomes $6^r \{22, 26, 31, 37, 44, 53, 64, 77, 92\}$. The base record exhaustively verified all 80,000,000,000 admissible starts through 10^{11} . We do not repeat that computation: conjugacy transfers those exact verified trajectories to $\{6^r y : y \leq 10^{11}, 5 \nmid y\} \subset \mathcal{D}_r$. Numerical examples in all three basins illustrate the transport explicitly. Global three-attractor convergence remains conjectural; the theorem establishes exact equivalence of the scaled and base dynamical problems, not a proof of unresolved global convergence.

1 Introduction

Residue-dependent affine maps followed by complete prime-adic reduction form a natural class of generalized Collatz and Syracuse systems; see Lagarias [1], Matthews and Watts [2], and Carnielli [3]. The author's preceding base study [4] defined the reduced $6k \pm \{1, 2\}/5$ map, established its elementary algebraic structure, exhibited two fixed points and a 9-cycle, and exhaustively tested every admissible start through 10^{11} .

The present paper asks a structural rather than computational question. If all four affine correction constants and the state space are simultaneously multiplied by 6^r , does a genuinely new dynamical system appear? We prove that it does not: for every integer $r \geq 0$, the scaled system is exactly

conjugate to the base system by multiplication by 6^r . This yields exact transport of complete trajectories, periodic structure, 5-adic valuation sequences, stopping times, basin membership, and normalized orbit geometry.

The 10^{11} numerical data quoted below belong to the base computation [4]. They are inherited through the conjugacy theorem and are not relabelled as independent 80-billion-start computations for each r .

2 The base reduced map

Let

$$\mathcal{D} = \{y \in \mathbb{Z}_{>0} : 5 \nmid y\}.$$

Define

$$c(y) = \begin{cases} -1, & y \equiv 1 \pmod{5}, \\ -2, & y \equiv 2 \pmod{5}, \\ +2, & y \equiv 3 \pmod{5}, \\ +1, & y \equiv 4 \pmod{5}, \end{cases} \quad M_0(y) = 6y + c(y),$$

and

$$T_0(y) = \frac{M_0(y)}{5^{\nu_5(M_0(y))}}. \quad (1)$$

Because $6 \equiv 1 \pmod{5}$, the selected correction makes $M_0(y)$ divisible by 5, and complete removal of its factors of 5 returns a member of $\mathcal{D}$.

The three observed attractors in the base finite verification are

$$1 \rightarrow 1, \quad 2 \rightarrow 2,$$

and

$$22 \rightarrow 26 \rightarrow 31 \rightarrow 37 \rightarrow 44 \rightarrow 53 \rightarrow 64 \rightarrow 77 \rightarrow 92 \rightarrow 22. \quad (2)$$

These are directly verified algebraically in the base paper [4].

3 The 6^r -scaled family

Fix $r \geq 0$ and define

$$\mathcal{D}_r = 6^r \mathcal{D} = \{6^r y : y \in \mathcal{D}\}. \quad (3)$$

Every $x \in \mathcal{D}_r$ has a unique representation $x = 6^r y$. Since

$$6^r \equiv 1 \pmod{5}, \quad (4)$$

we have $x \equiv y \pmod{5}$, so scaling preserves the branch residue.

Define

$$M_r(x) = 6x + 6^r c(x), \quad x \in \mathcal{D}_r, \quad (5)$$

where $c(x)$ is selected by the same residue rule modulo 5. Equivalently,

$$M_r(x) = \begin{cases} 6x - 6^r, & x \equiv 1 \pmod{5}, \\ 6x - 2 \cdot 6^r, & x \equiv 2 \pmod{5}, \\ 6x + 2 \cdot 6^r, & x \equiv 3 \pmod{5}, \\ 6x + 6^r, & x \equiv 4 \pmod{5}. \end{cases}$$

The compressed scaled map is

$$T_r(x) = \frac{M_r(x)}{5^{\nu_5(M_r(x))}}. \quad (6)$$

Equation (4) makes the appropriate branch numerator divisible by 5, so T_r is well-defined on $\mathcal{D}_r$.

4 Key 5-adic scaling identities

Lemma 4.1 (Affine scaling). For every $r \geq 0$ and $y \in \mathcal{D}$,

$$M_r(6^r y) = 6^r M_0(y). \quad (7)$$

Proof. Since $6^r y \equiv y \pmod{5}$, we have $c(6^r y) = c(y)$. Therefore

$$M_r(6^r y) = 6(6^r y) + 6^r c(y) = 6^r(6y + c(y)) = 6^r M_0(y). \quad \square$$

Lemma 4.2 (Valuation invariance). For every nonzero integer n and $r \geq 0$,

$$\nu_5(6^r n) = \nu_5(n). \quad (8)$$

Proof. Since $\gcd(6^r, 5) = 1$, the scaling factor contributes no factor of 5. $\square$

Consequently,

$$\nu_5(M_r(6^r y)) = \nu_5(M_0(y)). \quad (9)$$

Thus corresponding steps remove exactly the same power of 5.

5 Exact algebraic conjugacy

Define the bijection

$$\Phi_r : \mathcal{D} \rightarrow \mathcal{D}_r, \quad \Phi_r(y) = 6^r y.$$

Theorem 5.1 (One-step conjugacy). For every $r \geq 0$ and $y \in \mathcal{D}$,

$$T_r(6^r y) = 6^r T_0(y). \quad (10)$$

Equivalently,

$$T_r \circ \Phi_r = \Phi_r \circ T_0, \quad T_r = \Phi_r \circ T_0 \circ \Phi_r^{-1}.$$

Proof. Combine (7) and (9) in (6). $\square$

Theorem 5.2 (All iterates). For every $j \geq 0$,

$$T_r^j(6^r y) = 6^r T_0^j(y). \quad (11)$$

Proof. Induction on j using Theorem 5.1. $\square$

6 Three-attractor transport and cycle preservation

Theorem 6.1 (Cycle bijection). A tuple $(a_0, \dots, a_{L-1})$ is a cycle of T_0 if and only if $(6^r a_0, \dots, 6^r a_{L-1})$ is a cycle of T_r . Minimal cycle length is preserved.

Proof. Apply (11) with $j = L$ in both directions through the bijection Φ_r . A shorter period on one side would give the same shorter period on the other. $\square$

Hence the observed base attractors transport to

$$6^r \rightarrow 6^r, \quad 2 \cdot 6^r \rightarrow 2 \cdot 6^r, \quad (12)$$

and

$$22 \cdot 6^r \rightarrow 26 \cdot 6^r \rightarrow 31 \cdot 6^r \rightarrow 37 \cdot 6^r \rightarrow 44 \cdot 6^r \rightarrow 53 \cdot 6^r \rightarrow 64 \cdot 6^r \rightarrow 77 \cdot 6^r \rightarrow 92 \cdot 6^r \rightarrow 22 \cdot 6^r. \quad (13)$$

No periodic orbit can occur inside $\mathcal{D}_r$ unless dividing it by 6^r produces a base periodic orbit of the same minimal length. This statement does not assert that the three displayed base attractors are globally exhaustive; it says that scaled periodic structure is in bijection with base periodic structure.

At $r = 1$, the displayed attractors are 6, 12, and

$$132 \rightarrow 156 \rightarrow 186 \rightarrow 222 \rightarrow 264 \rightarrow 318 \rightarrow 384 \rightarrow 462 \rightarrow 552 \rightarrow 132.$$

At $r = 2$, they are 36, 72, and

$$792 \rightarrow 936 \rightarrow 1116 \rightarrow 1332 \rightarrow 1584 \rightarrow 1908 \rightarrow 2304 \rightarrow 2772 \rightarrow 3312 \rightarrow 792.$$

7 Numerical examples in all three basins

The base paper gives transparent examples entering each of the three observed basins. Conjugacy turns each one into an infinite family of exact examples.

Example A: capture by the scaled fixed point 6^r

The base trajectory from 9 is

$$9 \rightarrow 11 \rightarrow 13 \rightarrow 16 \rightarrow 19 \rightarrow 23 \rightarrow 28 \rightarrow 34 \rightarrow 41 \rightarrow 49 \rightarrow 59 \rightarrow 71 \rightarrow 17 \rightarrow 4 \rightarrow 1.$$

Therefore, for every $r \geq 0$,

$$9 \cdot 6^r \rightarrow 11 \cdot 6^r \rightarrow 13 \cdot 6^r \rightarrow \dots \rightarrow 4 \cdot 6^r \rightarrow 6^r.$$

For $r = 1$ this is

$$54 \rightarrow 66 \rightarrow 78 \rightarrow 96 \rightarrow 114 \rightarrow 138 \rightarrow 168 \rightarrow 204 \rightarrow 246 \rightarrow 294 \rightarrow 354 \rightarrow 426 \rightarrow 102 \rightarrow 24 \rightarrow 6.$$

For instance, $54 \equiv 4 \pmod{5}$ and the scaled branch gives $(6 \cdot 54 + 6)/5 = 66$. At $x = 102$, the numerator is $6 \cdot 102 - 2 \cdot 6 = 600 = 25 \cdot 24$, illustrating complete 5-adic reduction.

Example B: capture by the scaled fixed point $2 \cdot 6^r$

The base trajectory $24 \rightarrow 29 \rightarrow 7 \rightarrow 8 \rightarrow 2$ yields

$$24 \cdot 6^r \rightarrow 29 \cdot 6^r \rightarrow 7 \cdot 6^r \rightarrow 8 \cdot 6^r \rightarrow 2 \cdot 6^r.$$

At $r = 1$,

$$144 \rightarrow 174 \rightarrow 42 \rightarrow 48 \rightarrow 12.$$

Indeed $M_1(174) = 6 \cdot 174 + 6 = 1050 = 25 \cdot 42$, so the same valuation $\nu_5 = 2$ appearing at the corresponding base step is preserved.

Example C: capture by the scaled 9-cycle

The base trajectory

$$36 \rightarrow 43 \rightarrow 52 \rightarrow 62 \rightarrow 74 \rightarrow 89 \rightarrow 107 \rightarrow 128 \rightarrow 154 \rightarrow 37$$

enters the 9-cycle. Thus

$$36 \cdot 6^r \rightarrow 43 \cdot 6^r \rightarrow 52 \cdot 6^r \rightarrow \cdots \rightarrow 154 \cdot 6^r \rightarrow 37 \cdot 6^r$$

enters (13). At $r = 1$ the transient is

$$216 \rightarrow 258 \rightarrow 312 \rightarrow 372 \rightarrow 444 \rightarrow 534 \rightarrow 642 \rightarrow 768 \rightarrow 924 \rightarrow 222,$$

after which

$$222 \rightarrow 264 \rightarrow 318 \rightarrow 384 \rightarrow 462 \rightarrow 552 \rightarrow 132 \rightarrow 156 \rightarrow 186 \rightarrow 222.$$

The final transient step has $M_1(924) = 6 \cdot 924 + 6 = 5550 = 25 \cdot 222$, matching the base two-factor reduction from 154 to 37.

8 Cycle equation under scaling

For a scaled periodic orbit write

$$5^{a_i} x_{i+1} = 6x_i + 6^r c_i, \quad c_i \in \{-2, -1, 1, 2\}. \quad (14)$$

Putting $x_i = 6^r y_i$ and dividing by 6^r recovers exactly

$$5^{a_i} y_{i+1} = 6y_i + c_i.$$

Thus the scaled cycle equation contains no new Diophantine condition. If $A = \sum_{i=0}^{L-1} a_i$ and $S_j = \sum_{i=0}^{j-1} a_i$, successive substitution gives

$$(5^A - 6^L)x_0 = 6^r \sum_{j=0}^{L-1} 6^{L-1-j} c_j 5^{S_j}. \quad (15)$$

Dividing by 6^r is precisely the base cycle identity. This supplies a second direct algebraic view of cycle transport.

9 Orbit statistics preserved by conjugacy

Let $y_j = T_0^j(y)$ and $x_j = T_r^j(6^r y)$. Then $x_j = 6^r y_j$ for all j .

Transient/stopping time. First entry into the corresponding attractor occurs at the same iteration, so

$$\tau_r(6^r y) = \tau_0(y).$$

The complete valuation sequence is identical step by step, hence the accumulated number of removed factors of 5 before attractor entry is also identical.

Peaks. For every matched finite trajectory segment,

$$\max_j x_j = 6^r \max_j y_j,$$

while

$$\frac{x_j}{x_0} = \frac{y_j}{y_0}.$$

Thus absolute reduced peaks scale by 6^r and every normalized peak ratio is invariant.

Basins. If C is a base cycle, then

$$6^r y \in B_r(6^r C) \iff y \in B_0(C).$$

Therefore basin counts are exactly preserved on corresponding finite sets. Counts at the same absolute cutoff should not be conflated with counts on corresponding scaled sets.

10 Five-adic drift viewpoint

The base residue model uses

$$\Pr(\nu_5 = k) = \frac{4}{5^k}, \quad k \geq 1, \quad \mathbb{E}[\nu_5] = \frac{5}{4},$$

with heuristic mean logarithmic increment

$$\Delta \approx \log 6 - \frac{5}{4} \log 5 \approx -0.2200379213$$

and geometric typical multiplier $6/5^{5/4} \approx 0.802488366$ [4]. For matched states,

$$\frac{T_r(6^r y)}{6^r y} = \frac{T_0(y)}{y},$$

so normalized one-step ratios are exactly identical and the scaled family inherits the same heuristic drift model. As in the base work, this heuristic is not a proof of global convergence.

11 Finite verification inherited from the base record

The base computation [4] exhaustively tested every $1 \leq y \leq 10^{11}$ with $5 \nmid y$, a total of 80,000,000,000 starts. The reported full-cutoff basin counts were: The base run reported zero additional cycles,

Observed attractor	Count	Percentage
Fixed point 1	41,199,583,259	51.49947907%
Fixed point 2	11,613,268,146	14.51658518%
9-cycle (2)	27,187,148,595	33.98393574%
Total	80,000,000,000	100%

Table 1: Base-system finite verification through 10^{11} , inherited from the supplied base record [4].

zero unsigned-128 overflow events, and zero step-limit events. Its maximum stopping time was 755, at $y = 63,909,268,656$, and its largest reduced peak was

$$18,774,256,269,070,896,812,821,$$

from $y = 73,433,001,822$ [4].

For every fixed r , Theorem 5.2 therefore verifies algebraically the corresponding set

$$\{6^r y : 1 \leq y \leq 10^{11}, 5 \nmid y\} \subset \mathcal{D}_r. \quad (16)$$

It contains the same 80 billion starts and extends through absolute scale $6^r 10^{11}$ within the scaled domain. Basin counts and stopping times are identical. The 755-step record corresponds to scaled start

$$6^r \cdot 63,909,268,656,$$

and the largest reduced peak on the corresponding scaled set is

$$6^r \cdot 18,774,256,269,070,896,812,821,$$

from scaled start $6^r \cdot 73,433,001,822$.

These are exact consequences of conjugacy applied to the inherited base verification, not new measurements from independently rerunning the full search for each r .

12 Finite theorem and global conjecture

Finite transfer theorem. For every $r \geq 0$, every $x = 6^r y$ with $1 \leq y \leq 10^{11}$ and $5 \nmid y$ enters one of the two scaled fixed points in (12) or the scaled 9-cycle (13). This follows from the exhaustive base finite verification together with Theorem 5.2.

Conjecture (scaled three-attractor convergence). For every $r \geq 0$ and every $x \in \mathcal{D}_r$, the orbit of x under T_r eventually enters one of the two scaled fixed points or the scaled 9-cycle.

Theorem 12.1 (Equivalence of global conjectures). The scaled three-attractor conjecture holds for any one $r \geq 0$ if and only if the base three-attractor conjecture holds on $\mathcal{D}$. Consequently it holds for all $r \geq 0$ if and only if it holds for $r = 0$.

Proof. The bijective conjugacy transports every orbit in both directions. $\square$

Thus the parameter r creates no additional global convergence problem, but it also does not solve the base problem. Any undiscovered base cycle or unbounded base trajectory would have a scaled counterpart at every level.

13 What is proved and what is not

For every integer $r \geq 0$, the algebra proves: (i) $\mathcal{D}_r = 6^r \mathcal{D}$ is in bijection with the base domain; (ii) multiplication by 6^r preserves residues modulo 5; (iii) $M_r(6^r y) = 6^r M_0(y)$; (iv) the complete 5-adic valuation is preserved; (v) T_r and T_0 are exactly conjugate; (vi) all iterates correspond; (vii) cycles and minimal periods correspond bijectively; (viii) valuation sequences and transient lengths are preserved; (ix) absolute peaks scale by 6^r while normalized ratios are invariant; (x) basin membership transfers exactly; and (xi) the complete base verification through 10^{11} transfers to the corresponding set (16) through absolute scale $6^r 10^{11}$.

What is not proved is global convergence of every base orbit, absence of all additional base cycles, or absence of an unbounded base orbit. The exhaustive 10^{11} computation is finite. Exact conjugacy preserves this unresolved status rather than removing it.

14 Reproducibility archive

The accompanying archive contains this L^AT_EX source and compiled PDF, a Python verifier, a machine-readable JSON statement of the scaled rules, the inherited base 10^{11} summary, the supplied base-paper source as provenance material, a README, and SHA-256 checksums. The verifier checks finite samples of

$$M_r(6^r y) = 6^r M_0(y), \quad \nu_5(M_r(6^r y)) = \nu_5(M_0(y)), \quad T_r(6^r y) = 6^r T_0(y),$$

and several iterates. It is a consistency check, not the proof; the proof is the factorization in Lemma 4.1 together with $6^r \equiv 1 \pmod{5}$ and $\gcd(6^r, 5) = 1$.

15 Conclusion

Multiplying the state space and all four correction constants of the reduced $6k \pm \{1, 2\}/5$ map by 6^r produces an exact scaled copy of the base dynamics. The central identity $T_r(6^r y) = 6^r T_0(y)$ holds for every $r \geq 0$, so complete trajectories, valuation sequences, stopping times, basin membership, and cycle lengths are preserved exactly; only absolute state magnitudes and peaks acquire the factor 6^r .

Accordingly, the two observed base fixed points become 6^r and $2 \cdot 6^r$, and the observed 9-cycle becomes its 6^r multiple. The exhaustive base verification of 80 billion admissible starts through 10^{11} transfers without a new large computation to the corresponding scaled set through absolute scale $6^r 10^{11}$. The global convergence question is unchanged: proving the base three-attractor conjecture would prove it simultaneously for the entire scaled family, while any base counterexample would propagate through the same conjugacy.

References

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